

Chapter 1

Methodology

The simulations for FIV in the present study are carried out in the open-source CFD tool, OpenFOAM. In the following section, the governing equations and numerical schemes are discussed.

1.1 Flow Solver

The complex fluid flow field is numerically simulated by solving the non-dimensional Navier-Stokes equations governing the conservation of momentum and mass for a viscous, incompressible, Newtonian fluid. To effectively accommodate the moving boundary problem and prevent severe cell distortion during large mesh deformations around the oscillating body, the Arbitrary Lagrangian-Eulerian (ALE) description of kinematic motion is implemented. Unlike purely Eulerian methods where the grid remains static, or purely Lagrangian methods where the grid tracks material particles, the ALE formulation allows the computational grid to move independently of the fluid particle velocity. The resulting non-dimensional governing momentum and continuity equations for the fluid flow are formulated as follows:

$$\frac{\partial \mathbf{u}}{\partial t} + ((\mathbf{u} - \mathbf{u}_{mesh}) \cdot \nabla) \mathbf{u} = -\nabla p + \frac{1}{\text{Re}} (\nabla^2 \mathbf{u}), \quad (1.1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (1.2)$$

where $\mathbf{u}$, $\mathbf{u}_{mesh}$, and p denote the non-dimensionalized fluid flow velocity vector, the computational mesh velocity vector, and the fluid kinematic pressure, respectively. The non-dimensional Reynolds number is defined as $\text{Re} = \rho U_\infty D / \mu$, representing the ratio of inertial forces to viscous forces. To achieve the non-dimensional form of the governing equations, the major diameter of the elliptical cylinder D serves as the characteristic length scale, while the uniform free-stream velocity U_∞ serves as the reference velocity scale. Consequently, the non-dimensional temporal domain is defined by $t = \frac{t^* U_\infty}{D}$, where t^* denotes the physical, dimensional time.

1.2 Structure Motion Solver

The rigid-body structural dynamics of the elastically mounted elliptical cylinder, restricted to a single degree-of-freedom transverse to the oncoming flow, are governed

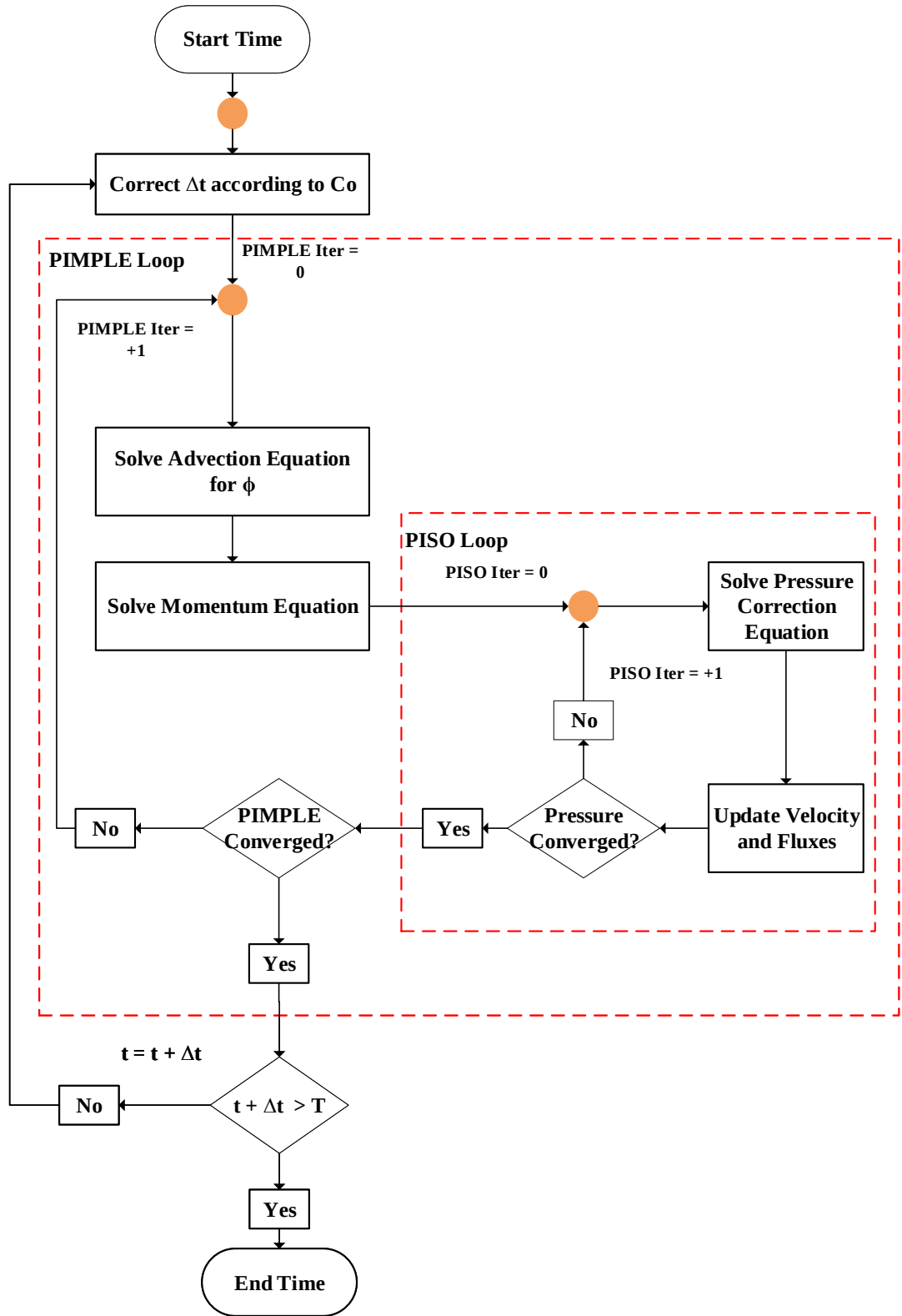


Figure 1.1: Flowchart of PIMPLE Algorithm

by a non-dimensional spring-mass-damper ordinary differential equation (Soti et al. [40]):

$$\ddot{Y} + \frac{4\pi\zeta}{U^*}\dot{Y} + \frac{4\pi^2}{(U^*)^2}Y = \frac{2}{\pi} \frac{C_l}{m^*} \frac{1}{AR}, \quad (1.3)$$

where $Y = y/D$, $\dot{Y}$, and $\ddot{Y}$ represent the non-dimensional transverse displacement, non-dimensional velocity, and non-dimensional acceleration of the cylinder center of mass, respectively, derived with respect to the non-dimensional time domain. The instantaneous fluid forces acting on the body are coupled via the non-dimensional lift coefficient, expressed as $C_l = F_y / \frac{1}{2}\rho U_\infty^2 D L_c$, where F_y is the integrated vertical hydrodynamic force.

The structural system's inertia is characterized by the mass ratio $m^* = m/(m_f)$, which relates the total structural moving mass m to the fluid mass displaced by the cylinder body $m_f = \pi\rho D^2 L_c (AR)/4$. The parameter AR denotes the structural aspect ratio of the elliptical geometry, defined by the ratio of the minor diameter to the major diameter ($AR = b/D$), and L_c represents the nominal spanwise length of the cylinder. The structural response is heavily driven by the reduced velocity parameter, $U^* = U_\infty/(f_n D)$, which scales the free-stream velocity against the system's structural natural frequency in a vacuum, defined as:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}, \quad (1.4)$$

where k denotes the linear structural spring stiffness. The structural damping ratio is given by $\zeta = c/c_c$, which relates the physical damping coefficient c to the critical damping limit of the system ($c_c = 4\pi m f_n$). In this specific investigation, the system is modeled as undamped, and thus ζ is explicitly set to zero ($\zeta = 0$).

To maintain tight kinematic coupling between the fluid forces and structural response, the ordinary differential equation 1.3 is integrated step-by-step using the second-order accurate, implicit Newmark- β numerical integration method. The foundational kinematic approximations of the Newmark- β algorithm mapping state variables from time step n to $n + 1$ are formulated as:

- **Velocity $\dot{y}_{n+1}$:**

$$\dot{y}_{n+1} = \dot{y}_n + \Delta t \ddot{y}_\alpha$$

where the averaged acceleration vector $\ddot{y}_\alpha$ is parameterized as:

$$\ddot{y}_\alpha = (1 - \alpha)\ddot{y}_n + \alpha\ddot{y}_{n+1}.$$

- **Displacement y_{n+1} :**

$$y_{n+1} = y_n + \Delta t \dot{y}_n + \frac{1}{2} \Delta t^2 \ddot{y}_\beta$$

where the weighted acceleration vector $\ddot{y}_\beta$ is parameterized as:

$$\ddot{y}_\beta = (1 - 2\beta)\ddot{y}_n + 2\beta\ddot{y}_{n+1}.$$

By enforcing dynamic equilibrium, these discrete kinematic expressions are directly substituted into the structural equation of motion evaluated at the future time step $n + 1$:

$$m\ddot{y}_{n+1} + c\dot{y}_{n+1} + ky_{n+1} = F(t)_{n+1} \quad (1.5)$$

By isolating the unknown future kinematics through algebraic substitution, the underlying differential system is re-cast into an equivalent linear algebraic system. Written in a compact matrix form, it is solved directly for the updated displacement vector u_{n+1} :

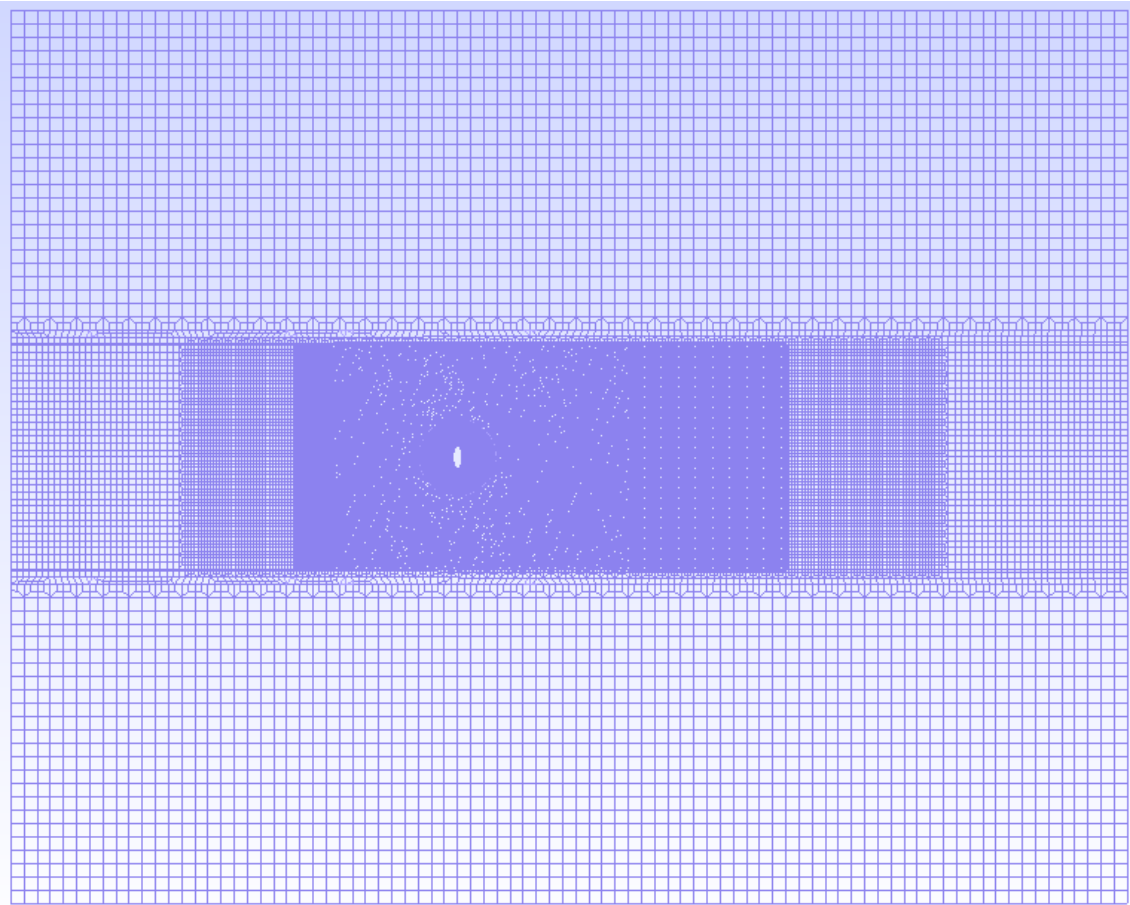
$$[K] \cdot u_{n+1} = [F] \quad (1.6)$$

where $[K]$ represents the effective dynamic stiffness matrix incorporating mass, damping, and stiffness contributions, and $[F]$ represents the effective dynamic load vector acting on the structure. The standard execution parameters for the Newmark- β scheme are selected as $\alpha = 0.5$ and $\beta = 0.25$, which ensures that the integration routine behaves as an unconditionally stable, second-order accurate constant average acceleration operator.

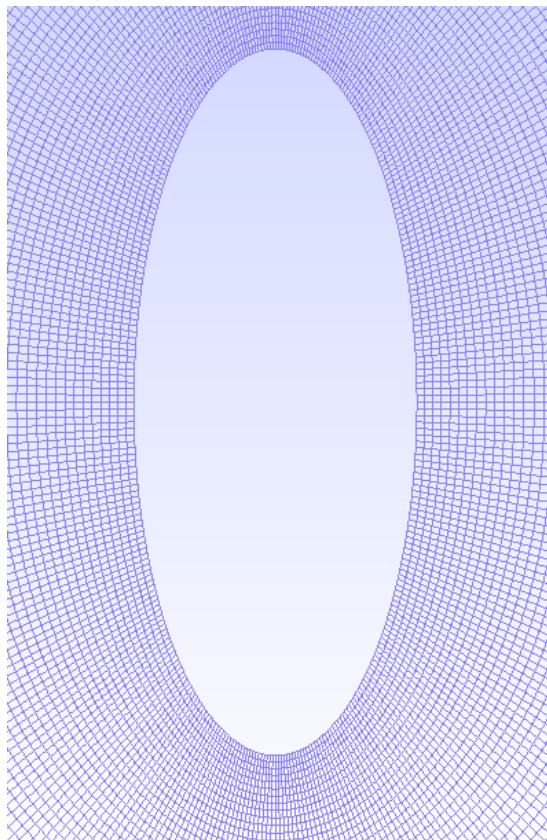
1.3 Mesh Strategy

The computational mesh utilized in this study is generated using the open-source three-dimensional finite element grid generator Gmsh [41]. To achieve a balance between spatial accuracy and computational efficiency, a hybrid structured/unstructured grid generation strategy is meticulously deployed. This approach exploits the respective strengths of different topological configurations across distinct zones of the computational domain, determined by the localized flow physics, expected velocity gradients, and proximity to the solid boundaries. The geometric complexity introduced by the curved boundaries of the elliptical cylinder within a rectangular domain is effectively managed using an unstructured grid strategy in the transition regions. This ensures a smooth geometric conformation to the cylinder wall without inducing high cell skewness or sudden aspect ratio jumps. The global domain is systematically partitioned into three distinct concentric and adjacent topological zones:

1. **Near-Wall Boundary Layer Zone (Inner Core):** A fine, non-uniform structured grid (O-grid/C-grid) is enforced immediately around the elliptical cylinder. Elements are clustered heavily toward the wall using a geometric progression stretch factor to ensure the non-dimensional wall distance (y^+) resolves the viscous sublayer directly. This high-density resolution is critical for capturing high gradients, skin friction, and vortex-shedding frequencies.
2. **Intermediate Transition Zone:** A medium-sized unstructured grid encapsulates the inner core, serving as a topological buffer that smoothly transitions from the curvilinear inner mesh to the rectilinear far-field layout. This zone utilizes unstructured elements with controlled size propagation to allow mesh density to diffuse outward, preventing numerical reflections and artificial damping of shed vortices in the near-wake.



(a) Mesh of the Total Domain



(b) Mesh around the Cylinder at $AR = 0.40$

Figure 1.2: Computational mesh layout

3. **Far-Field Zone (Outer Domain):** The outer region extending to the domain boundaries is discretized using a non-uniform structured grid. Because flow gradients decay significantly far from the cylinder, cell sizes grow progressively toward the external boundaries using a directional stretching law. This minimizes the total grid count and optimizes computational efficiency within OpenFOAM.

This multi-tiered, hybrid meshing methodology ensures that critical wake physics, shear-layer instabilities, and FIV responses are captured with high geometric fidelity, while successfully minimizing unnecessary memory consumption and keeping the overall computational cost optimized.

1.4 Mathematical Formulation for Structural Vibration and Force Response

The vibration amplitude is defined as:

$$A^* = \frac{y_{\max} - y_{\min}}{2} \quad (1.7)$$

where $y_{\max}$ and $y_{\min}$ represent the extrema of the cylinder displacements.

A Fast Fourier Transform (FFT) is applied to the transient lift and displacement signals to extract the frequency characteristics, while the Hilbert Transform is utilized to determine the instantaneous phase relationship.

The RMS lift coefficients and the mean drag coefficients are computed using the formulas given as:

$$C'_L = \sqrt{\frac{1}{N} \sum_{i=1}^N (C_L^i)^2}, \quad (1.8)$$

$$\overline{C}_D = \frac{1}{N} \sum_{i=1}^N C_D^i. \quad (1.9)$$

where N denotes the number of observations recorded during a particular period of the system's periodic or quasi-periodic behavior; C_L^i and C_D^i denote the non-dimensional lift and drag forces, respectively, for the time step i .

Chapter 2

FIV of Two Elliptical Cylinders in Tandem Arrangement at $Re = 100$

2.1 Problem Definition

The effect of gap ratio on the flow-induced vibration (FIV) of two rigidly coupled elliptical cylinders in tandem arrangement is numerically investigated. Both the cylinders undergo the same displacement in the transverse direction. The flow is fairly two-dimensional at a mass ratio of $m^* = 10$ and a constant Reynolds number of $Re = 100$ [42]. The gap ratio (L^*) in this case is the normalized center-to-center distance between the cylinders ($L^* = \frac{L}{D}$), and m^* is the ratio of the total oscillating mass m and the mass of the displaced fluid ($m^* = \frac{2m}{\pi \rho D^2 AR}$). To investigate the effects of spacing, tandem gap ratios are investigated across a range of $1.5 \leq L^* \leq 6.0$, a selection based on established literature on the flow-induced vibration (FIV) of rigidly coupled cylinders. Following the classification established by Badri et al. [20], these configurations are categorized into three distinct regimes: small ($L^* = 1.5, 2.0$), medium ($L^* = 3.0, 4.0$), and large ($L^* = 5.0, 6.0$). For each discrete gap ratio, numerical simulations are systematically conducted across a reduced velocity spectrum of $2.0 \leq U^* \leq 14.0$. A previous mesh independence study led to the selection of a final mesh size of 0.01 near the cylinder walls.

2.2 FIV Response Characteristics

2.2.1 Structural Vibration Response

The left columns of Figs. 2.2, 2.3, and 2.4 illustrate the vibration amplitude response (A^*) and the phase difference (ϕ) between the cylinder displacement and the total lift force as a function of reduced velocity (U^*). A distinct phase jump from 0° to 180° is observed, signaling the transition from the initial branch to the lower branch [43]. Correspondingly, the right columns present the relevant frequency components: vortex shedding frequency (f_v), oscillation frequency (f), structural natural frequency (f_n), and the stationary cylinder shedding frequency (f_s). Notably, the significance of ϕ is contingent upon the synchronization of f_v and f ; any divergence between these frequencies renders ϕ time-dependent and

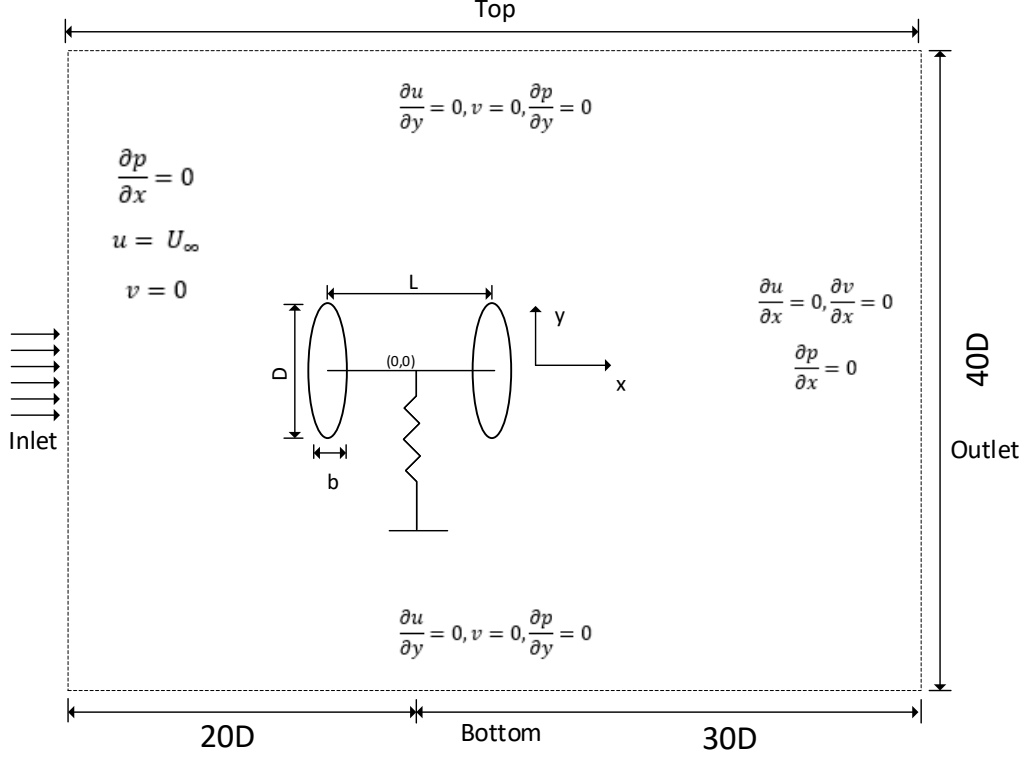


Figure 2.1: Computational Domain

physically ambiguous. In such cases, large deviations from the 0° and 180° will yield misleading interpretations regarding the energy transfer between the fluid and the structure.

2.2.2 Force Response

Figs. 2.5–2.7 detail the behavior of the lift and drag forces acting on the cylinder system. In the left columns, the root-mean-square (RMS) lift coefficients for the upstream cylinder ($C'_{L_{UC}}$), downstream cylinder ($C'_{L_{DC}}$), and the coupled system ($C'_{L_{Total}}$) are plotted, where the latter represents the algebraic sum of both lift forces. Similarly, the right columns present the mean drag coefficients of the upstream ($\overline{C}_{D_{UC}}$) and downstream ($\overline{C}_{D_{DC}}$) cylinders alongside the total mean drag coefficient ($\overline{C}_{D_{Total}}$).

2.2.3 Flow Patterns

This section analyzes the final state of the wake patterns characterizing the dual-cylinder configuration. Prior literature on tandem cylinder arrangements has established that the gap flow is predominantly governed by reattachment and co-shedding regimes. To systematically evaluate these flow physics, eight distinct wake modes are classified.

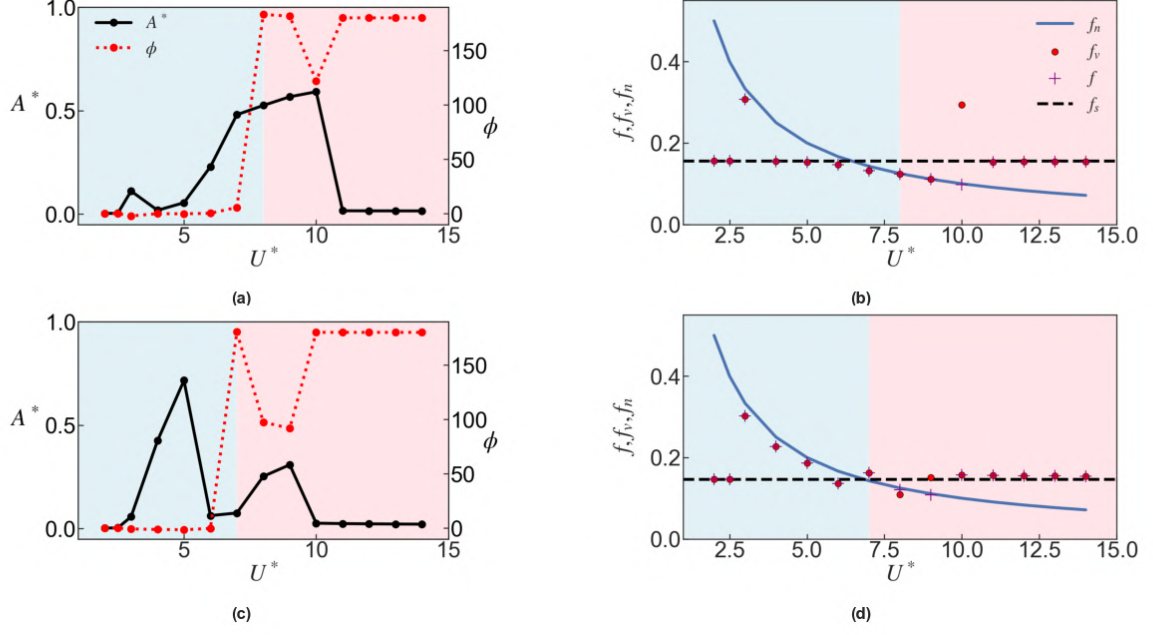


Figure 2.2: For $L^* = 1.5$: (a) Amplitude and phase response, and (b) Frequency response. For $L^* = 2.0$: (c) Amplitude and phase response, and (d) Frequency response. Light blue denotes the initial branch, and light pink denotes the lower branch

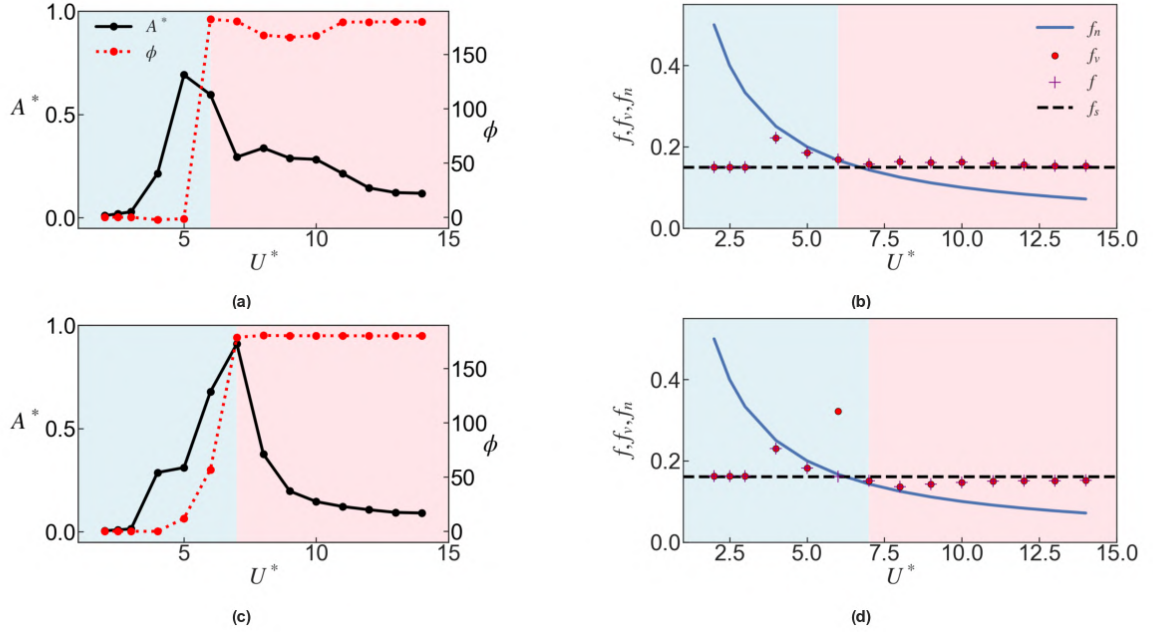


Figure 2.3: For $L^* = 3.0$: (a) Amplitude and phase response, and (b) Frequency response. For $L^* = 4.0$: (c) Amplitude and phase response, and (d) Frequency response. Light blue denotes the initial branch, and light pink denotes the lower branch

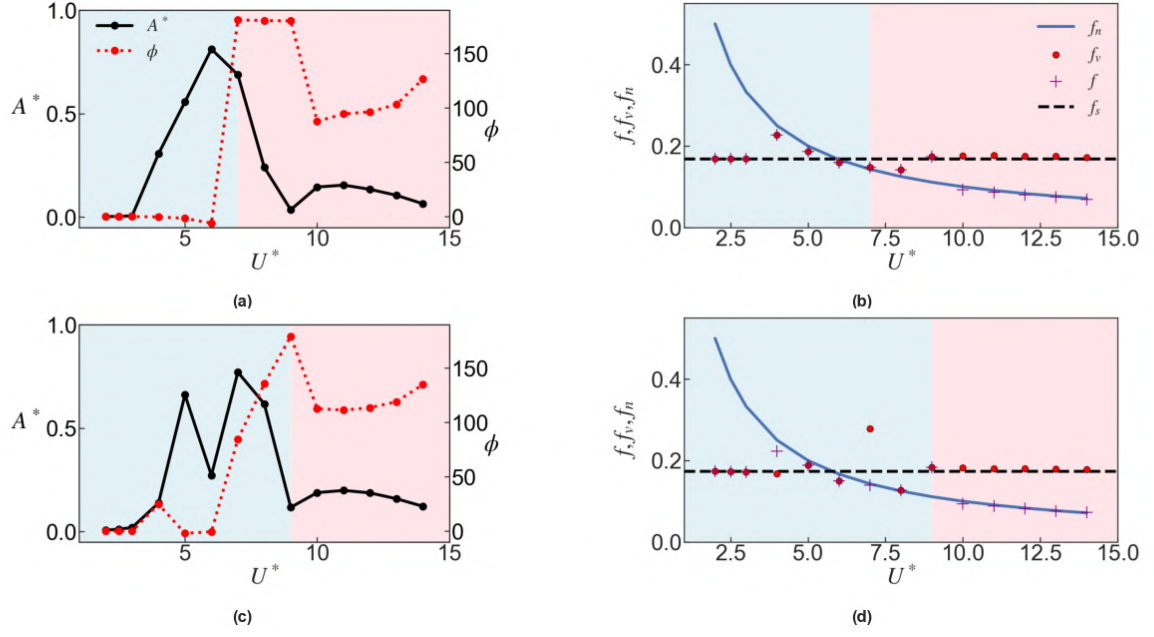


Figure 2.4: For $L^* = 5.0$: (a) Amplitude and phase response, and (b) Frequency response. For $L^* = 6.0$: (c) Amplitude and phase response, and (d) Frequency response. Light blue denotes the initial branch, and light pink denotes the lower branch

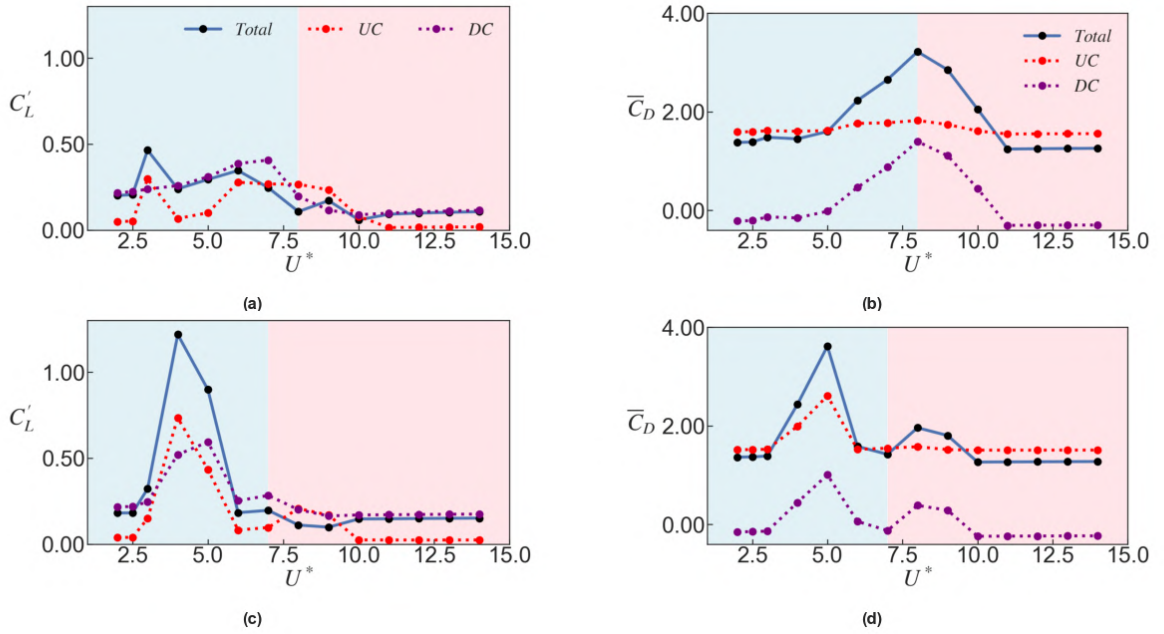


Figure 2.5: For $L^* = 1.5$: (a) RMS lift coefficient, and (b) Mean drag coefficient. For $L^* = 2.0$: (c) RMS lift coefficient, and (d) Mean drag coefficient. Light blue denotes the initial branch, and light pink denotes the lower branch

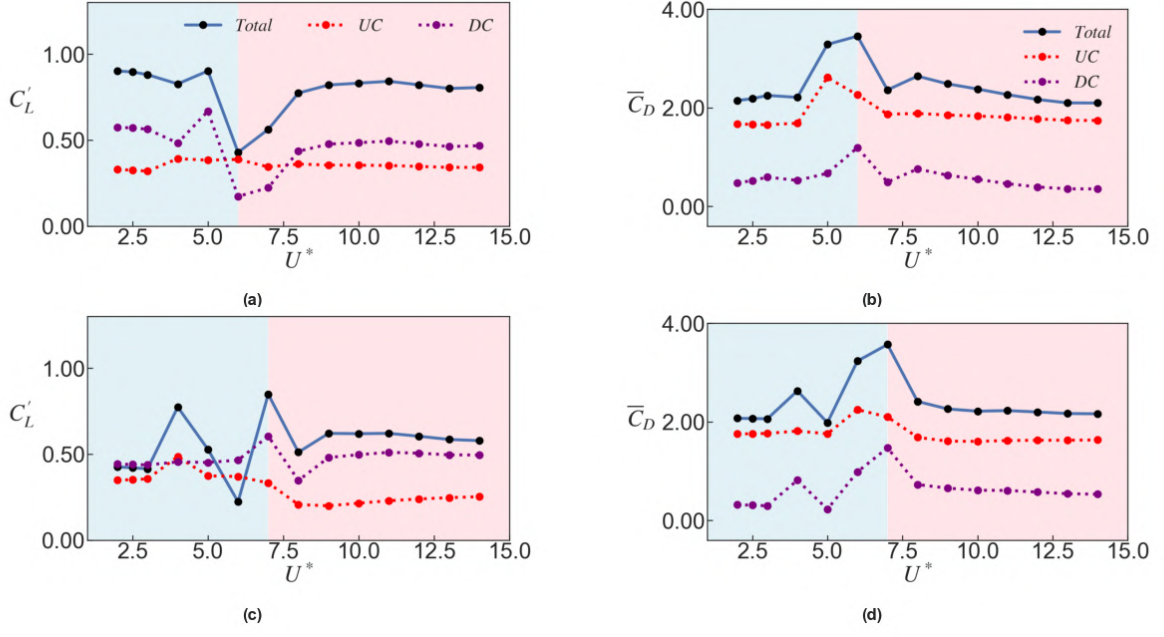


Figure 2.6: For $L^* = 3.0$: (a) RMS lift coefficient, and (b) Mean drag coefficient. For $L^* = 4.0$: (c) RMS lift coefficient, and (d) Mean drag coefficient. Light blue denotes the initial branch, and light pink denotes the lower branch

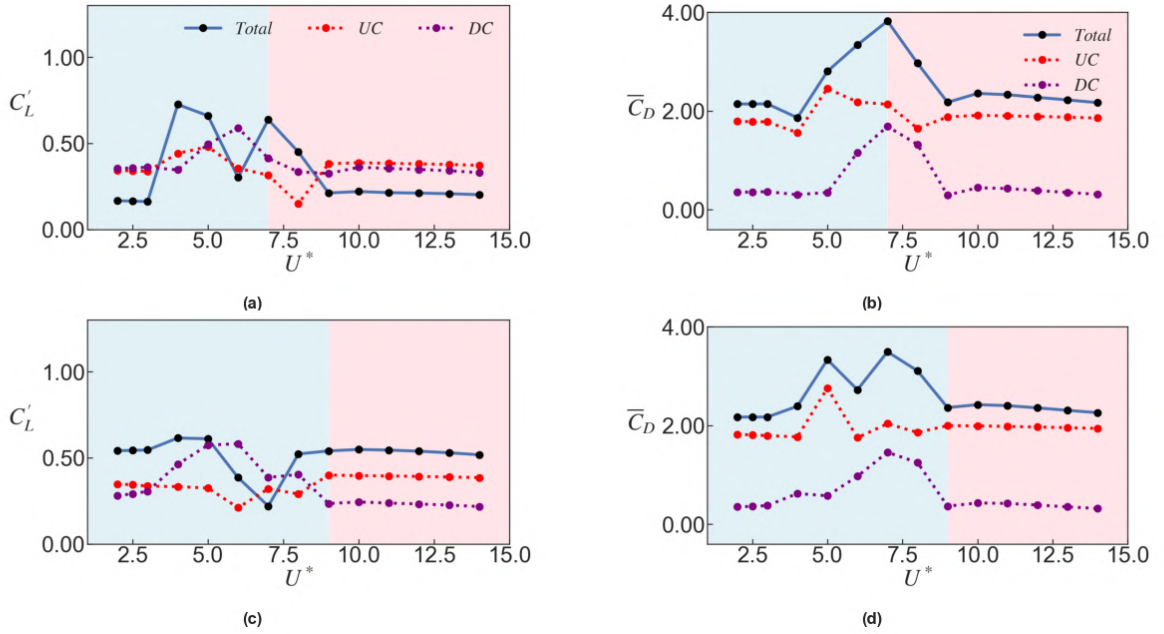


Figure 2.7: For $L^* = 5.0$: (a) RMS lift coefficient, and (b) Mean drag coefficient. For $L^* = 6.0$: (c) RMS lift coefficient, and (d) Mean drag coefficient. Light blue denotes the initial branch, and light pink denotes the lower branch

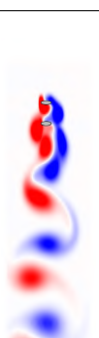
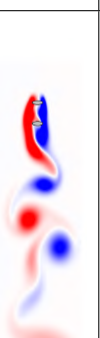
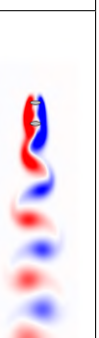
U^*	3.0	6.0	7.0	8.0	9.0	10.0	11.0
$L^* = 1.5$							
	I	I	IV	IV	I	I	I
U^*	3.0	4.0	5.0	6.0	8.0	9.0	10.0
$L^* = 2.0$							
	I	II	VI	I	IV	IV	I

Figure 2.8: The flow regimes and wake patterns for small gap ratios. The vortex scale is between -1(blue) and 1(red). The corresponding vortex pattern modes are defined in the rows below the wake patterns

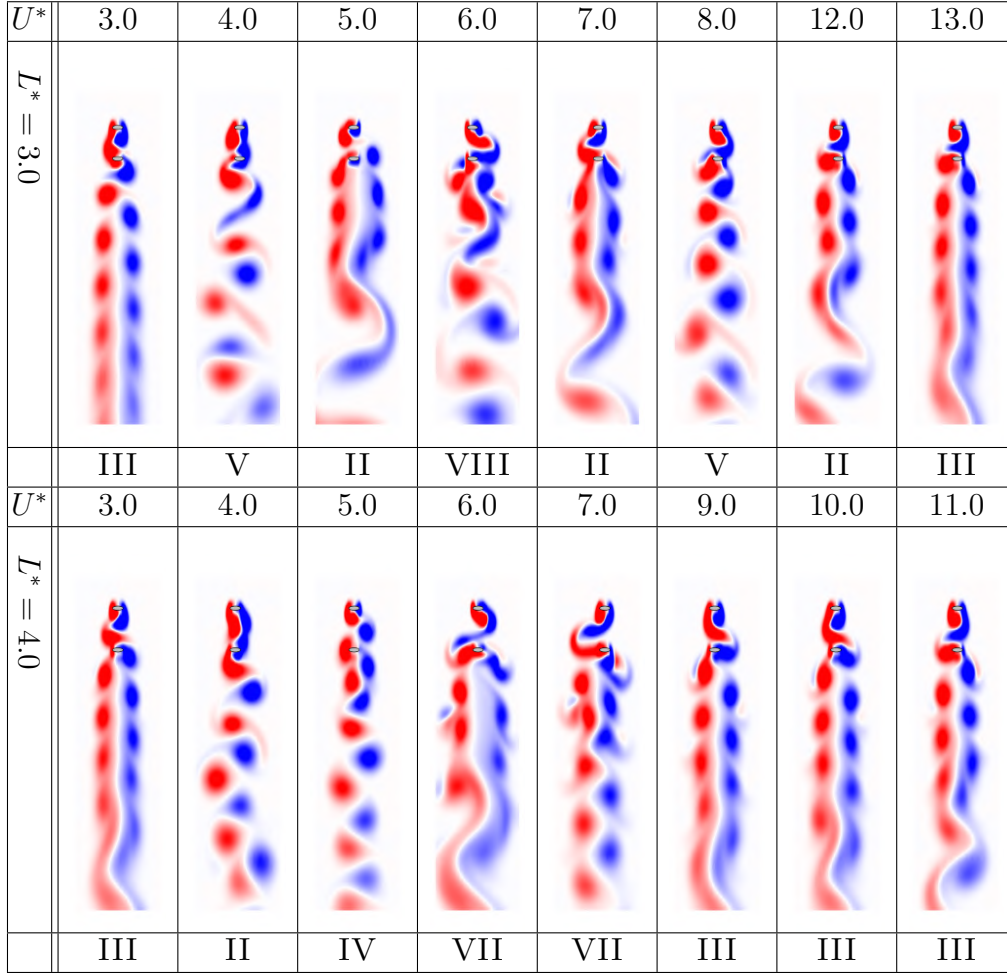


Figure 2.9: The flow regimes and wake patterns for medium gap ratios. The vortex scale is between -1(blue) and 1(red). The corresponding vortex pattern modes are defined in the rows below the wake patterns

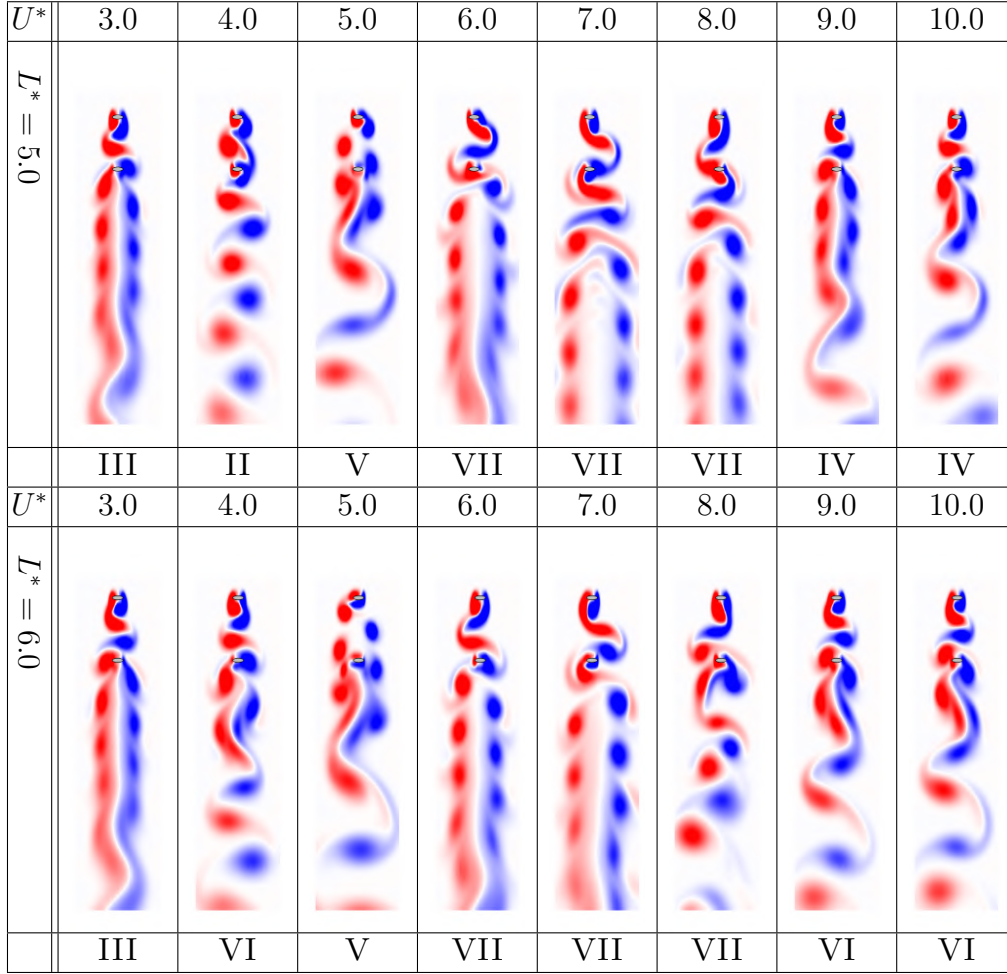


Figure 2.10: The flow regimes and wake patterns for large gap ratios. The vortex scale is between -1(blue) and 1(red). The corresponding vortex pattern modes are defined in the rows below the wake patterns

Chapter 3

Spanwise Parameter Study of Stationary Elliptical Cylinder at $Re = 500$

3.1 Problem Definition

The effect of spanwise length and spanwise grid resolution on the flow over a stationary elliptical cylinder at high Reynolds number is numerically investigated. At $Re = 500$, 3D DNS studies are performed for the study of the turbulent wake downstream of the cylinder. For resolving the smallest eddies in turbulent fluid flow, Kolmogorov scale is used to choose the mesh size near the cylinder wall. Kolmogorov estimates that the ratio of the largest to the smallest length scales is proportional to $Re^{3/4}$, where Re is based on the largest length scale [45]. Hence the finest mesh size is chosen to be 0.009 for $Re = 500$ in the two-dimensional plane of the domain. Studies are conducted for different spanwise length ratios (Z/D) where Z is the spanwise length, at a fixed spanwise grid resolution (dz) of $0.1D$ (the spanwise grid resolution chosen is consistent with existing literature for circular cylinders) for finding the Strouhal number ($St = \frac{f_v D}{U_\infty}$) and force coefficients (C'_L and $\overline{C}_D$) at different values of Z/D . At the optimum spanwise length ratio, the effects of spanwise grid resolution on the same variables are investigated. The variations in the flow pattern for different values of Z/D are also studied to identify the evolution of 3D wake structures with spanwise dimension.

3.2 Three-dimensional Wake Characteristics

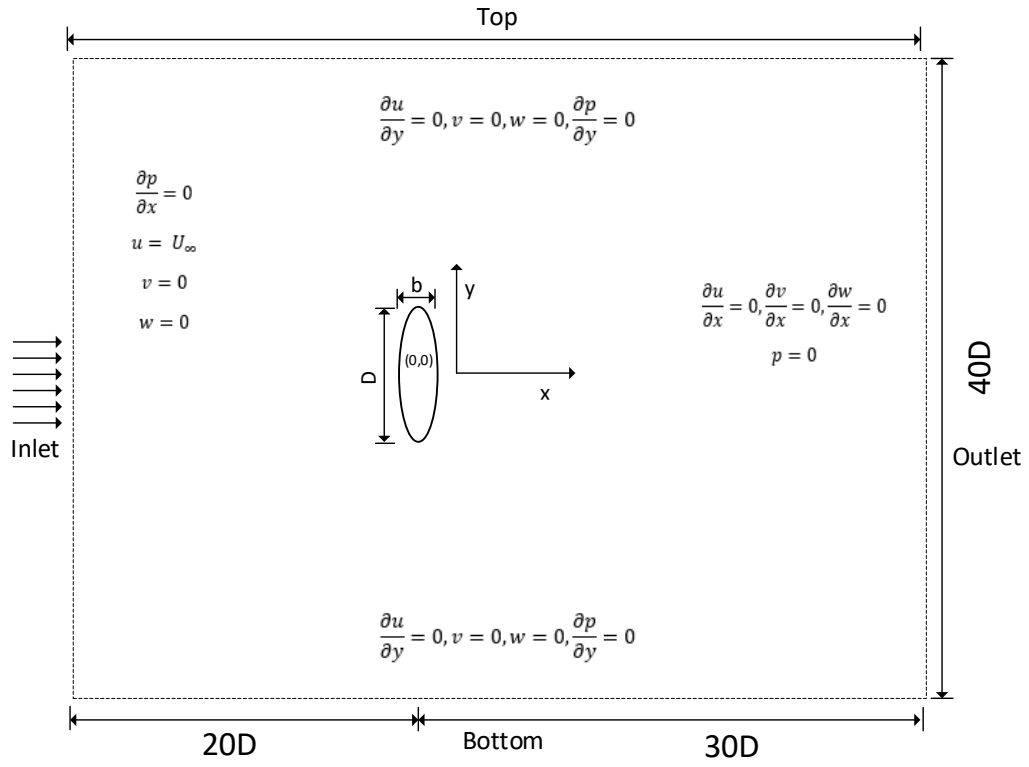


Figure 3.1: 2D Plane of Computational Domain

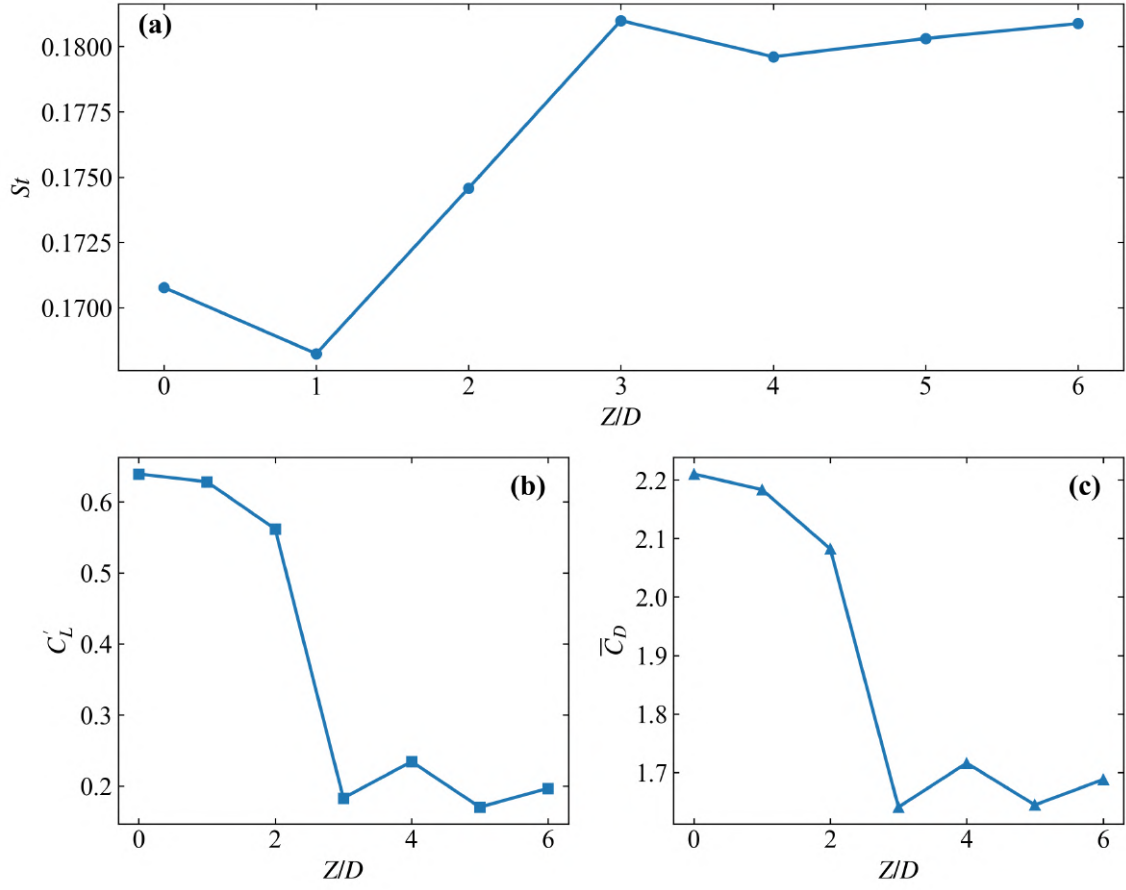


Figure 3.2: Variation of (a) Strouhal number, (b) RMS Lift Coefficient, and (c) Mean Drag Coefficient with Spanwise Length Ratio for AR = 0.4 Elliptical Cylinder

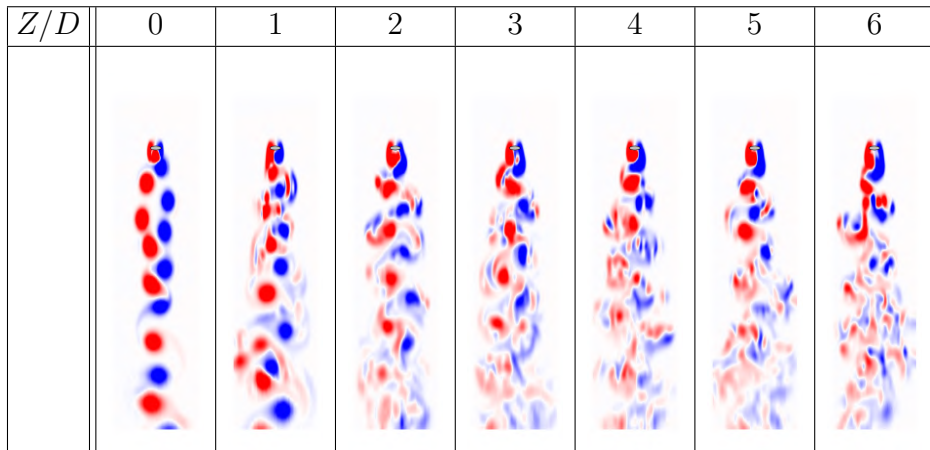


Figure 3.3: The wake patterns at the mid-plane in the spanwise direction. The vortex scale is between -1(blue) and 1(red)

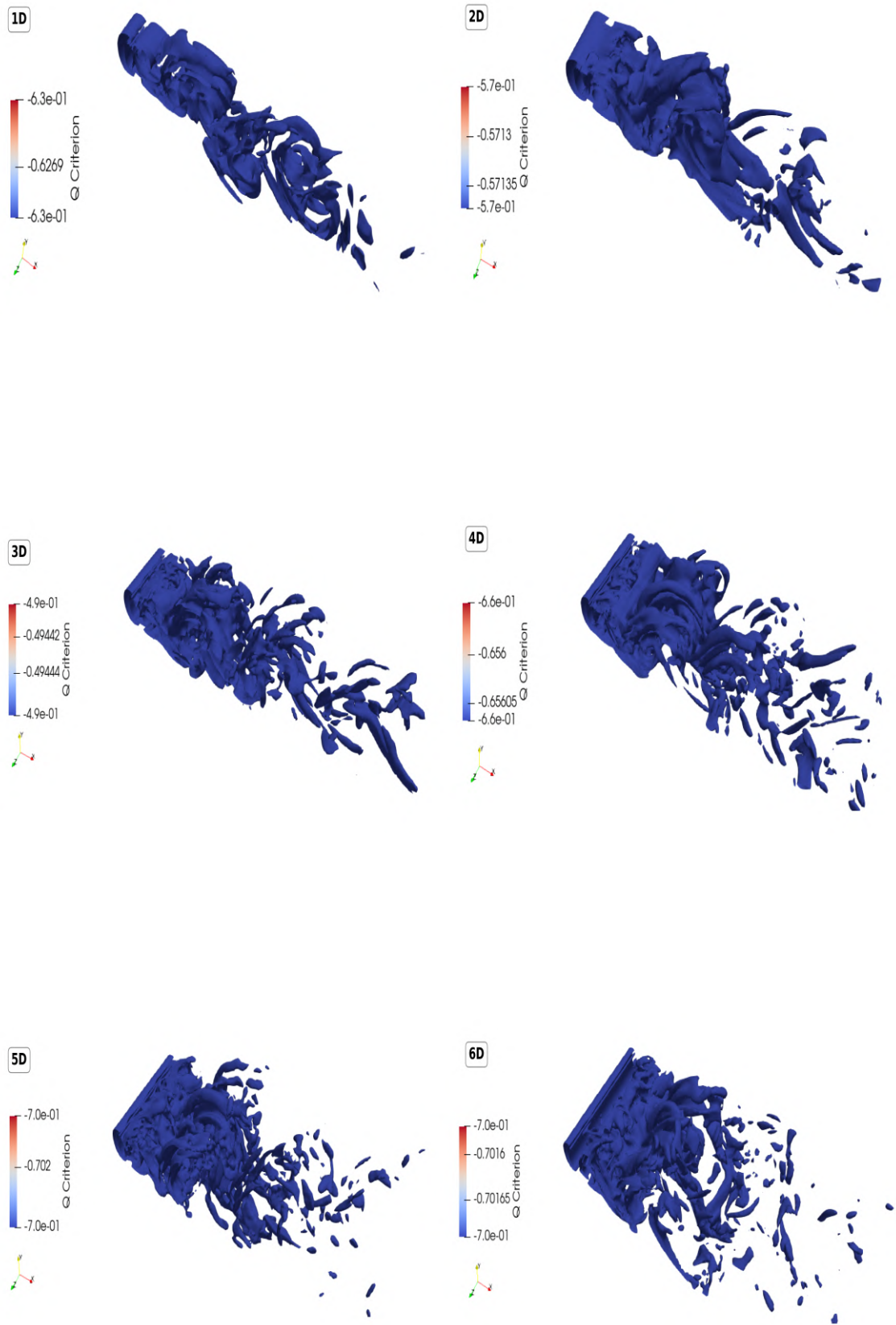


Figure 3.4: Q-criterion Iso-surface Plots at Different Spanwise Lengths

Chapter 4

FIV of Single Elliptical Cylinder at $\text{Re} = 500$

4.1 Problem Definition

The effect of high Reynolds number on the flow-induced vibration (FIV) of single elliptical cylinder is numerically investigated. 3D DNS studies are performed at a mass ratio of $m^* = 10$ and a constant Reynolds number of $\text{Re} = 500$. The mass ratio (m^*) in this case is the ratio of the total oscillating mass m and the mass of the displaced fluid ($m^* = \frac{4m}{\pi\rho D^2 AR}$). Numerical simulations are systematically conducted across a reduced velocity range of $1.0 \leq U^* \leq 14.0$ using the optimum spanwise length and spanwise grid resolution for flow over stationary elliptical cylinder of the same AR. A comparative analysis between 2D and 3D simulations was conducted to validate the performance of the 3D numerical framework in resolving turbulent wake-induced FIV phenomena.

4.2 FIV Response Characteristics

4.2.1 Structural Vibration Response

The left column of Fig. 4.2 delineates the non-dimensional cross-flow vibration amplitude response (A^*) and the corresponding phase difference (ϕ) between the structural displacement and the transverse hydrodynamic lift force coefficient, plotted as a function of the reduced velocity (U^*). Complementing these kinematic and force-matching metrics, the right column presents a comprehensive spectral mapping of the fundamental frequencies governing the coupled fluid-structure interaction (FSI) system: the dynamic vortex shedding frequency (f_v), the primary structural oscillation frequency (f), the structural natural frequency (f_n), and the baseline Strouhal shedding frequency of an equivalent stationary cylinder (f_s).

4.2.2 Force Response

Fig. 4.3 details the variation of the hydrodynamic force coefficients acting on the oscillating cylinder across the designated parameter space. Specifically, the top

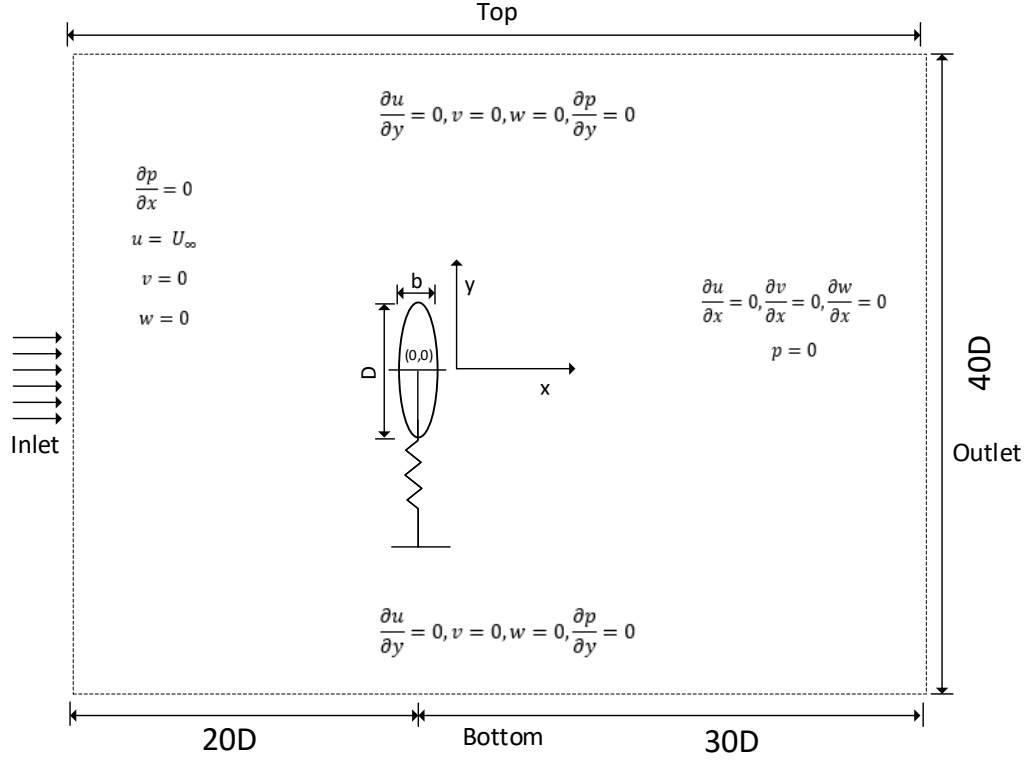


Figure 4.1: 2D Plane of Computational Domain

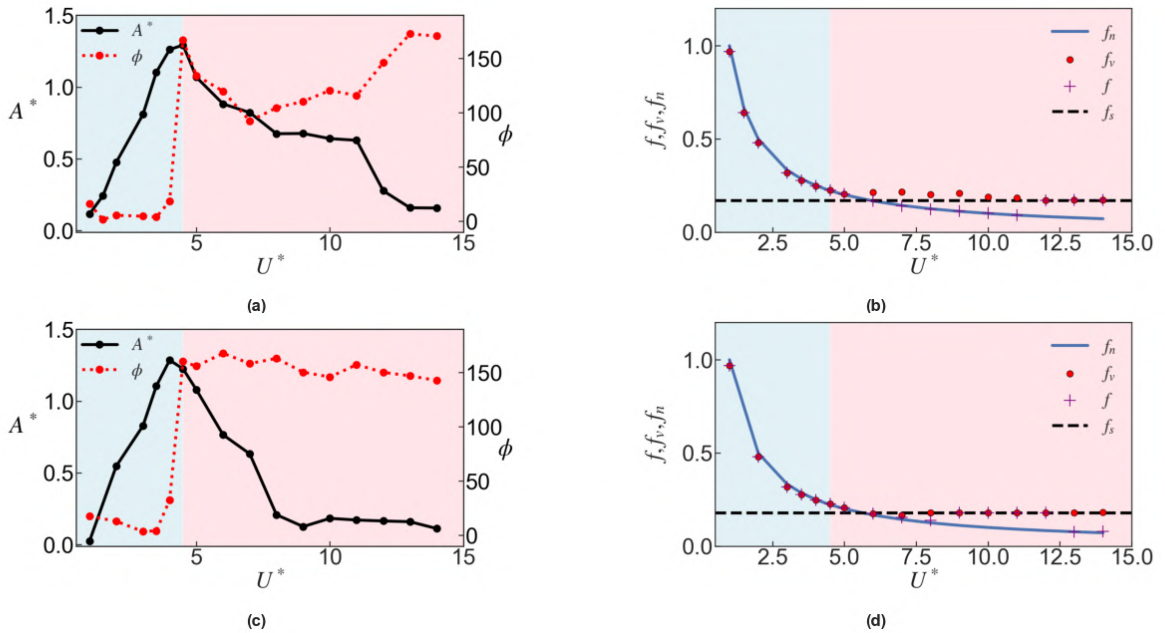


Figure 4.2: For 2D studies: (a) Amplitude and phase response, and (b) Frequency response. For 3D studies: (c) Amplitude and phase response, and (d) Frequency response. Light blue denotes the initial branch, and light pink denotes the lower branch

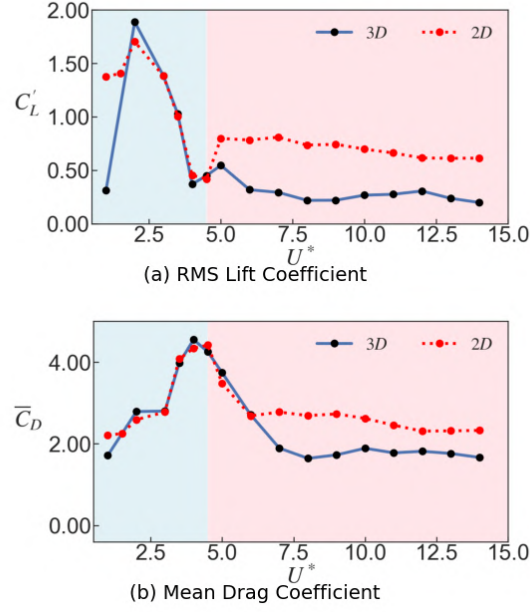


Figure 4.3: Comparison of Force Coefficients for 3D and 2D studies. Light blue denotes initial branch and light pink denotes lower branch

row delineates the root-mean-square (RMS) lift coefficient (C'_L) for both the three-dimensional (3D) and two-dimensional (2D) simulations, tracking the temporal intensity of the transverse fluid forces. Complementing this, the bottom row presents the time-averaged mean drag coefficient ($\bar{C}_D$), which acts as a primary indicator of the streamwise aerodynamic resistance and suction pressure in the near wake.